

Folder 04: Data Extraction & Synthesis

1. Content Overview

This folder provides the structured datasets extracted from the **110 primary studies** included in this SLR. Unlike traditional manual extraction, this study employed a **Hybrid AI-Human Extraction Workflow** to ensure technical depth and accuracy across Four specific Research Questions (RQs):

- **RQ1 (Dataset Characteristics & Adoption):** Identification of the core attributes and most frequently utilized datasets (e.g., CERT, LANL, Enron) in machine learning-based insider threat detection.
- **RQ2 (Dataset Evaluation & Applicability):** Analysis of the strengths, inherent limitations, and specific use cases across different data environments, including synthetic, real-world, multi-source, and domain-specific datasets.
- **RQ3 (Data Preprocessing Techniques):** Examination of the specific methods applied to transform raw data and unstructured logs into behavioral representations suitable for machine learning models (e.g., data cleaning, normalization, and formatting).
- **RQ4 (Data Related Challenges):** Evaluation of underlying data-centric issues that impact the reproducibility, reliability, and broader generalizability of insider threat detection systems.

2. Extraction Methodology

To ensure the highest level of academic integrity, the following pipeline was utilized:

1. **AI-Assisted Synthesis:** We utilized **Google Gemini** to perform initial technical extractions from the full-text PDFs of all 110 studies.
2. **Manual Verification:** Every data point generated by the AI was cross-referenced against the original source manuscript by both authors to ensure that data synthesis was performed in duplicate.
3. **Cross-Checking:** Technical parameters (e.g., specific window sizes for temporal features or normalization formulas) were manually verified to eliminate "hallucinations" and ensure scientific accuracy.

3. Document Descriptions

File Name	Description
04_01_Master_Extraction_Table.pdf	The Master Table: Contains the raw extraction data for all included studies. Each row represents a study, with columns mapping to dataset characteristics (RQ1), dataset evaluations and limitations (RQ2), preprocessing pipelines (RQ3), and reported data challenges (RQ4).

04_02_RQ1_Dataset_Matrix.pdf	Focused Dataset Report: Dedicated to RQ1 and RQ2. This file provides a tabular breakdown of every dataset, detailing its characteristics, source type (synthetic, real-world, multi-source, domain-specific), strengths, and inherent limitations.
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4. Data Validation Note

All extractions have been validated for consistency by two reviewers. The qualitative synthesis provided in the manuscript (Section V) is directly derived from the structured data contained in these files.